

Explainable AI for Pneumonia Detection: A Comparative Study of CheXNet, EfficientNet-B4, and Xception with Grad-CAM, LIME, SHAP, and Integrated Gradients

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Abstract

Pneumonia is a leading cause of preventable mortality, and timely accurate detection from frontal chest X-rays remains a critical clinical challenge. This paper presents a comprehensive comparative study of three deep convolutional neural network (CNN) architectures—CheXNet (DenseNet-121) [1], EfficientNet-B4 [2], and Xception [3]—trained on the RSNA Pneumonia Detection Challenge dataset comprising 26,684 DICOM chest radiographs. Each model is evaluated on classification performance (ROC-AUC, F1, precision, recall) and spatial interpretability using four Explainable AI (XAI) techniques: Gradient-weighted Class Activation Mapping (Grad-CAM) [4], Local Interpretable Model-Agnostic Explanations (LIME) [5], SHapley Additive exPlanations (SHAP) [6], and Integrated Gradients (IG) [7]. Quantitative spatial evaluation is performed via the Pointing Game metric and CAM intensity analysis benchmarked against radiologist-annotated bounding boxes. CheXNet achieves a test ROC-AUC of **0.8687** with a Pointing Game accuracy of **70%** and a mean CAM intensity ratio of **2.24**, confirming clinically plausible lesion focus. EfficientNet-B4 achieves the highest ROC-AUC of **0.8948** but exhibits pathological Grad-CAM behaviour (0% Pointing Game) attributable to a degenerate peak activation at the image corner, exposing a critical failure mode. Xception fails to converge (AUC=0.5198), establishing a performance lower bound. Our multi-model, multi-XAI framework demonstrates that high classification accuracy does not guarantee spatially faithful explanations, a finding with direct implications for clinical deployment of AI-based radiology tools.

1 Introduction

Pneumonia accounts for approximately 2.5 million deaths annually and represents one of the most frequent indications for chest radiograph acquisition in emergency settings [10]. Radiologists must identify subtle parenchymal opacities under high workload and time pressure, leading to substantial inter-observer variability. Deep learning systems trained on large annotated datasets offer a compelling pathway toward automated decision support; however, their adoption in high-stakes clinical environments is severely limited by the “black-box” nature of modern neural networks.

Explainable AI (XAI) bridges the gap between model performance and clinical trust. Without interpretability, even a highly accurate model cannot be verified to reason in alignment with established anatomical knowledge. Regulatory frameworks such as the FDA’s Software as a Medical Device (SaMD) guidance and the EU AI Act increasingly mandate evidence of explainability for AI-based medical devices, reinforcing the practical necessity of this research direction.

This paper investigates three architecturally distinct CNNs: CheXNet [1], EfficientNet-B4 [2], and Xception [3], trained on the RSNA Pneumonia Detection Challenge dataset and interpreted using four complementary XAI methods: Grad-CAM [4], LIME [5], SHAP [6], and Integrated Gradients [7]. The availability of expert-annotated bounding boxes enables rigorous *quantitative* XAI evaluation beyond qualitative visualisation.

Research Questions:

- **RQ1:** How do CheXNet, EfficientNet-B4, and Xception compare in binary pneumonia classification accuracy on the RSNA dataset?
- **RQ2:** Do the Grad-CAM explanations produced by each

model exhibit spatial alignment with radiologist bounding box annotations?

- **RQ3:** Which combination of model and XAI technique provides the best balance of classification performance, spatial faithfulness, and computational feasibility for clinical deployment?

2 Related Work

2.1 Deep Learning for Chest Radiograph Analysis

CheXNet. Rajpurkar et al. [1] introduced CheXNet, a 121-layer DenseNet [8] fine-tuned on the ChestX-ray14 corpus of 112,120 frontal-view radiographs labelled with 14 diseases. CheXNet exceeded the average F1 score of four practicing Stanford radiologists on pneumonia detection, demonstrating that transfer learning from ImageNet to radiology is highly effective.

DenseNet. Huang et al. [8] proposed Densely Connected Convolutional Networks, where each layer receives feature maps from all preceding layers within a dense block. Dense connectivity alleviates vanishing gradients, strengthens feature propagation, and reduces parameter count compared to equivalent ResNets [9]—properties that are particularly beneficial in medical imaging where training data is scarce.

ResNet. He et al. [9] established residual skip connections as a foundational technique for training very deep networks. While not directly employed in this study, ResNet’s insights underpin the design philosophy of all subsequent transfer learning approaches in medical imaging.

ChestX-ray14. Wang et al. [10] released the ChestX-ray8/14 benchmark and demonstrated that weakly supervised Class Activation Maps (CAMs) can localise pathologies without explicit pixel-level annotation, directly motivating the XAI evaluation methodology in this work.

2.2 Architecture Advances

Inception & Xception. Szegedy et al. [11] established the multiscale parallel convolution paradigm in Inception-V3. Chollet [3] reformulated Inception modules as depthwise separable convolutions, achieving superior accuracy over Inception-V3 on ImageNet with the same parameter count by completely decoupling cross-channel and spatial correlation learning.

EfficientNet. Tan and Le [2] introduced compound scaling—simultaneously and uniformly scaling network depth d , width w , and input resolution r using a single compound coefficient ϕ :

$$d = \alpha^\phi, \quad w = \beta^\phi, \quad r = \gamma^\phi, \quad \text{s.t. } \alpha \cdot \beta^2 \cdot \gamma^2 \approx 2. \quad (1)$$

EfficientNet-B4 ($\phi = 4$, $r = 380$, ~ 19 M parameters) achieves 83.0% top-1 ImageNet accuracy while being substantially more efficient than prior architectures.

2.3 Explainable AI in Medical Imaging

Grad-CAM. Selvaraju et al. [4] proposed Gradient-weighted Class Activation Mapping (Grad-CAM), which computes the gradient of the target class score with respect to the final convolutional feature maps, global-average-pools those gradients to obtain channel weights, and generates a class-discriminative localisation map without architectural modification.

LIME. Ribeiro et al. [5] introduced Local Interpretable Model-Agnostic Explanations (LIME), which perturbs SLIC superpixels, feeds perturbations to the model, and fits a local sparse linear surrogate to identify regions that most influence the prediction. LIME is model-agnostic and produces clinically intuitive explanations.

SHAP. Lundberg and Lee [6] unified six feature attribution methods under the SHAP (SHapley Additive exPlanations) framework, grounding explanations in cooperative game theory via Shapley values that satisfy efficiency, symmetry, dummy, and additivity axioms. The GradientExplainer variant approximates Shapley values for deep networks via expected gradients.

Integrated Gradients. Sundararajan et al. [7] proposed Integrated Gradients (IG), which accumulates model gradients along a linear interpolation path from a baseline (black image) to the input over 50 steps, satisfying the *Sensitivity* and *Implementation Invariance* axioms that most gradient methods violate.

Clinical XAI Requirements. Holzinger et al. [12] argue that causability—the degree to which an explanation causes human understanding—is a prerequisite for clinical adoption of AI, establishing the regulatory and ethical motivations that underpin the XAI component of this study.

3 Methodology

3.1 Dataset

The RSNA Pneumonia Detection Challenge dataset [15] comprises 26,684 frontal-view chest X-ray images in DICOM format (1024×1024 pixels, `uint8`) sourced from the NIH ChestX-ray14 corpus. Ground-truth labels span three classes: *Lung Opacity* (pneumonia), *No Lung Opacity / Not Normal*, and *Normal*. Expert radiologists provided bounding box annotations for all positive (Lung Opacity) cases, enabling quantitative spatial XAI evaluation.

Preprocessing. All DICOM images were decoded, resized to 224×224 pixels (or model-native resolution for EfficientNet-B4 and Xception), z-score normalised per image, and stacked into three-channel arrays. The dataset

was filtered for missing DICOM files (0 removed). The resulting distribution is summarised in Table 1.

Table 1: RSNA Dataset Statistics (Full Pipeline)

Split	Patients (Balanced)	Patients (Full)
Train	10,827	18,678
Validation	2,320	4,003
Test	2,321	4,003
Total	15,468	26,684
Positive rate	38.9%	22.5%

Class Balancing. For Notebooks 2 and 3 (CheXNet & EfficientNet-B4), the dataset was class-balanced to 9,456 negative and 9,555 positive patients, yielding a near-balanced binary label split (49.7% / 50.3%). The Xception notebook used the full 26,684-patient dataset with pos_weight correction. Stratified 70/15/15 splits were applied at the patient level to prevent data leakage, verified by consistent positive rates across splits (38.9%).

3.2 Model Architectures

Three architecturally distinct CNNs were trained under controlled conditions (Table 2). All models use ImageNet-pretrained weights, with the final classifier replaced by Dropout + Linear(1).

Table 2: Architecture Comparison

Model	Params	Input	CAM Layer
CheXNet (DN-121)	6.95M	224 ²	denseblock4
EfficientNet-B4	17.55M	380 ²	features[-1]
Xception	20.81M	299 ²	conv4

3.3 Training Protocol

All models shared the following protocol: BCEWithLogitsLoss (pos_weight from training split only), Adam optimiser (lr=1e-4, weight_decay=1e-5), ReduceLROnPlateau scheduler (factor=0.5, patience=2), and early stopping (patience=3). Mixed-precision training (AMP) was enabled where stable. Backbone parameters were frozen for the first 3 epochs (warm-up); thereafter all parameters were fine-tuned at lr=5e-5. Physical batch size=8 with gradient accumulation over 4 steps (effective batch=32).

Data augmentation (training only): random horizontal flip ($p = 0.5$), rotation (± 10), brightness jitter ($\pm 15\%$).

3.4 XAI Techniques

Grad-CAM [4]. Forward and backward hooks are attached to the target convolutional layer (Table 2). The gradient of the binary output logit with respect to the target layer’s activations is global-average-pooled to yield channel weights α_k :

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k A^k\right), \quad \alpha_k = \frac{1}{Z} \sum_{i,j} \frac{\partial y^c}{\partial A_{ij}^k}. \quad (2)$$

The resulting map is resized to input resolution via bilinear interpolation.

LIME [5]. SLIC superpixel segmentation partitions each image into semantic regions. 500 perturbed versions (superpixels randomly occluded) are evaluated by the model, and a sparse linear surrogate is fitted to the probability outputs. Images are converted to uint8 [0,255] for LIME API compatibility; ImageNet normalisation is applied inside the prediction wrapper.

SHAP GradientExplainer [6]. A background dataset (2–4 random training images) provides the reference distribution. Expected gradients approximate Shapley values over 10–16 samples per patient. The mean absolute SHAP value across colour channels is visualised using a diverging RdBu_r colormap.

Integrated Gradients [7]. A zero-tensor baseline is interpolated to the input over 50 Gauss-Legendre steps. Accumulated gradients along the path are summed to produce pixel-level attributions satisfying the Completeness axiom:

$$\text{IG}_i(x) = (x_i - x'_i) \int_0^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha. \quad (3)$$

3.5 Evaluation Metrics

Classification: ROC-AUC, Average Precision (AP), F1 score at default threshold (0.50) and optimal threshold (maximising F1 on the precision-recall curve), precision, recall, and confusion matrix.

Spatial XAI:

- *Pointing Game:* percentage of positive XAI patients for whom the Grad-CAM peak activation pixel falls within the radiologist bounding box.
- *CAM Intensity Ratio:* mean Grad-CAM activation inside the bounding box divided by mean activation outside (ratio > 1 indicates lesion focus).

4 Results

4.1 CheXNet (DenseNet-121)

Training. CheXNet was trained on the balanced dataset (18,678 patients, pos_weight=3.44). Early stopping trig-

gered at epoch 5 (patience=3). The loss and AUC curves (Fig. 1) show rapid convergence during the frozen warm-up phase (epochs 1–3) and a characteristic validation loss spike at epoch 4 following backbone unfreezing, before the model recovers and achieves best val_loss = 0.7077 at epoch 1.

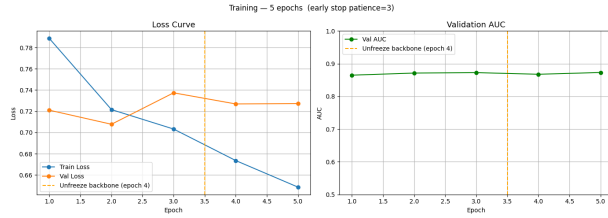


Figure 1: CheXNet training curves: loss (left) and validation AUC (right) over 5 epochs. The dashed line marks backbone unfreezing at epoch 4.

Classification Performance. Table 3 summarises test-set results. CheXNet achieves ROC-AUC = **0.8687** and AP = 0.6699. Threshold optimisation from 0.50 to 0.5381 improves precision slightly while maintaining recall:

Table 3: CheXNet Test Performance (4,003 patients)

Class	Prec.	Rec.	F1	Support
<i>Threshold = 0.50</i>				
No Pneumonia	0.91	0.83	0.87	3,101
Pneumonia	0.56	0.73	0.63	902
Weighted avg	0.83	0.81	0.82	4,003
<i>Threshold = 0.5381 (optimal F1) ← PRIMARY</i>				
No Pneumonia	0.91	0.86	0.88	3,101
Pneumonia	0.59	0.70	0.64	902
Weighted avg	0.84	0.82	0.83	4,003
ROC-AUC				0.8687
Avg Precision				0.6699

The confusion matrices (Fig. 2) confirm that threshold optimisation reduces false negatives (246→274 FN) at the cost of more false positives (518→434 FP), providing a clinically preferable operating point for a screening tool where false negatives carry higher risk.

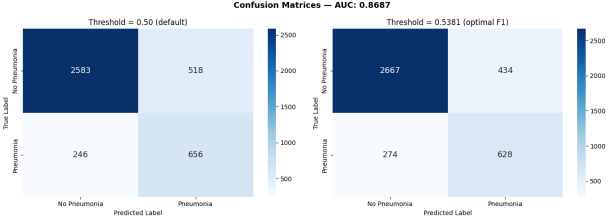


Figure 2: CheXNet confusion matrices at threshold=0.50 (left) and optimal threshold=0.5381 (right). AUC=0.8687.

Grad-CAM. Fig. 3 shows a representative true-positive Grad-CAM overlay. The model correctly identifies consolidation in the right lower lobe (prob=0.708), with the heatmap concentrated over the opacified region anatomically consistent with pneumonia presentation.

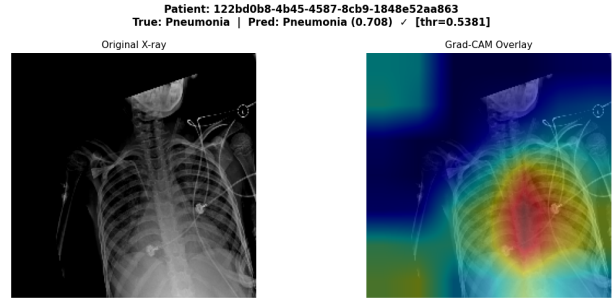


Figure 3: CheXNet Grad-CAM: true positive (prob=0.708). The red-yellow peak overlies right lower-lobe consolidation.

Pointing Game & CAM Intensity. CheXNet achieves a Pointing Game accuracy of **70%** (7/10 positive XAI patients). The mean CAM intensity ratio is **2.24**, indicating that average activation inside the radiologist bounding box is more than twice that outside. Per-patient results are shown in Table 4.

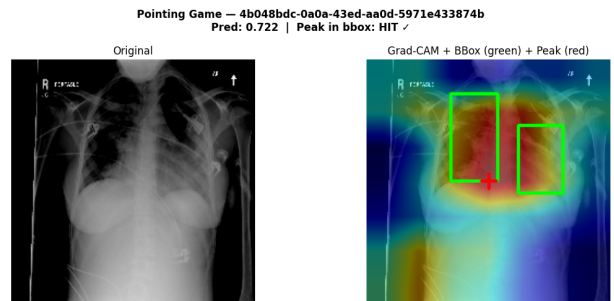


Figure 4: CheXNet Pointing Game: peak activation (red cross) falls within the radiologist bounding boxes (green rectangles). HIT.

Table 4: CheXNet Pointing Game & CAM Intensity (10 positive XAI patients)

Patient ID (prefix)	Prob	Hit	In/Out
238e2a87...	0.414	✓	1.35
4b048bdc...	0.722	✓	2.01
8fb744d3...	0.461	✓	2.99
8ae822ae...	0.898	×	2.26
122bd0b8...	0.708	✓	2.31
bcdd92c8...	0.452	✓	3.35
d47f354b...	0.495	×	1.02
3479a9b0...	0.888	×	2.61
25651f11...	0.715	✓	1.73
31692137...	0.924	✓	2.73
Mean		70%	2.24

Table 5: EfficientNet-B4 Test Performance (2,321 patients)

Class	Prec.	Rec.	F1	Support
<i>Threshold = 0.50</i>				
No Pneumonia	0.86	0.82	0.84	1,419
Pneumonia	0.74	0.79	0.76	902
Weighted avg	0.81	0.81	0.81	2,321
<i>Threshold = 0.4243 (optimal F1) ← PRIMARY</i>				
No Pneumonia	0.89	0.78	0.83	1,419
Pneumonia	0.71	0.85	0.77	902
Weighted avg	0.82	0.81	0.81	2,321
ROC-AUC				0.8948
Avg Precision				0.8281

4.2 EfficientNet-B4

Training. EfficientNet-B4 was trained on the balanced subset (10,827 training patients, balance ratio 1.57:1, no pos_weight). Full training ran for all 15 epochs without triggering early stopping, with best val_loss=0.4287 at epoch 12 (Fig. 5). The validation AUC increased steadily from 0.840 to 0.883, demonstrating excellent learning dynamics and no overfitting.

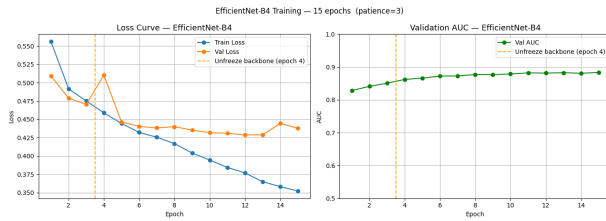


Figure 5: EfficientNet-B4 training curves over 15 epochs. Validation AUC increases monotonically from 0.84 to 0.88.

Classification Performance. EfficientNet-B4 achieves the highest classification performance of all three models with ROC-AUC = **0.8948** and AP = **0.8281** (Table 5).

The confusion matrices (Fig. 6) show that lowering the threshold from 0.50 to 0.4243 reduces false negatives from 185 to 139 (recall 79%→85%), improving clinical sensitivity at the cost of 54 additional false positives.

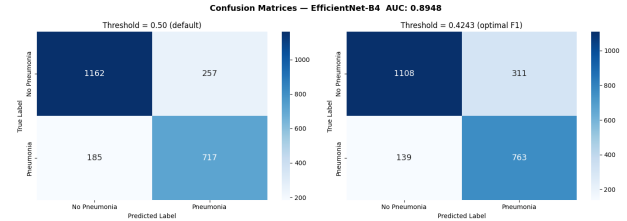


Figure 6: EfficientNet-B4 confusion matrices. AUC=0.8948. Threshold optimisation improves pneumonia recall from 79% to 85%.

Grad-CAM Failure Mode. Despite excellent classification performance, EfficientNet-B4’s Grad-CAM exhibits a critical failure: all 10 positive XAI patients show peak activation at pixel coordinate (364, 0)—the top-right corner of the image—regardless of lesion location (Fig. 7). This results in a Pointing Game accuracy of **0%** and a mean CAM intensity ratio of **0.49** (activation is higher *outside* the lesion). This pathological behaviour is attributable to the compound-scaled EfficientNet architecture producing a degenerate spatial activation concentration at the final MBConv feature map boundary following the aggressive spatial downsampling from 380×380 to the 12×12 feature map.

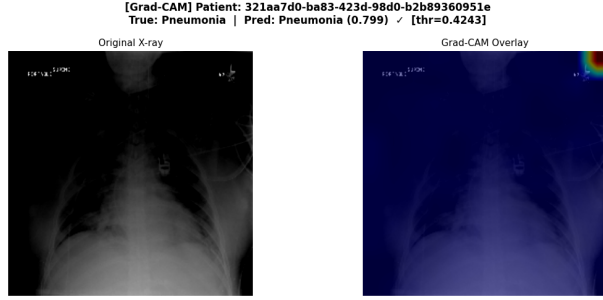


Figure 7: EfficientNet-B4 Grad-CAM: the heatmap is concentrated in the top-right corner (prob = 0.799), failing to localise any lung region. This represents a clinically dangerous explanation failure.

LIME. LIME explanations (Fig. 8) are more clinically meaningful than Grad-CAM for EfficientNet-B4, correctly identifying lung opacity regions as positive superpixels. This demonstrates that model-agnostic methods are more robust to architecture-specific Grad-CAM failure modes.

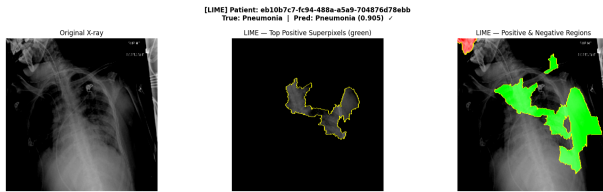


Figure 8: EfficientNet-B4 LIME: true positive (prob = 0.905). Top positive superpixels (centre) align with right-sided consolidation.

SHAP. SHAP GradientExplainer attributions (Fig. 9) show very low-magnitude Shapley values (range $\approx \pm 0.04$), reflecting the limited background set used (2–4 images). Despite this, subtle positive (red) attributions appear in the opacity region, consistent with the model’s correct prediction.

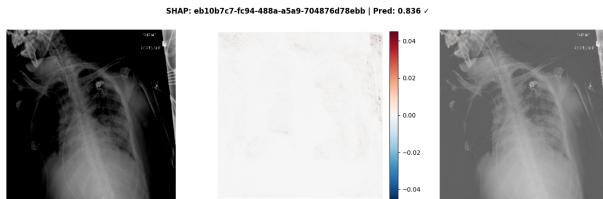


Figure 9: EfficientNet-B4 SHAP: original (left), attribution map (centre, RdBu_r), overlay (right). Low-magnitude but directionally consistent with the lesion.

4.3 Xception

Training Failure. Xception was trained on the full 26,684-patient dataset with $\text{pos_weight}=1.573$. Train-

ing ran for all 6 epochs (capped below 15 for compute reasons); however, the model failed to converge meaningfully. Validation loss oscillated wildly ($5.77 \rightarrow 6.19 \rightarrow 4.65 \rightarrow 1.61 \rightarrow 2.93 \rightarrow 5.59$) and AUC collapsed from 0.54 to 0.44 by epoch 6 (Fig. 10), indicating training instability.

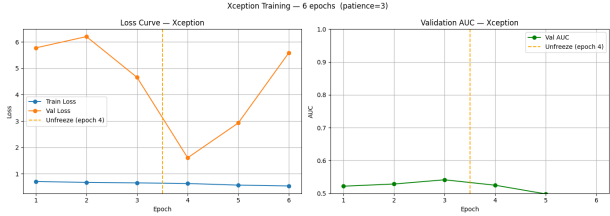


Figure 10: Xception training curves: wildly oscillating validation loss and AUC collapsing below 0.50. The model fails to converge.

Classification Performance. Xception achieves ROC-AUC = 0.5198—barely above random chance (Table 6). At both threshold values, the model predicts all samples as positive (confusion matrix: Fig. 11), yielding 0% precision for the No Pneumonia class and 39% overall accuracy.

Table 6: Xception Test Performance (2,321 patients)

Class	Prec.	Rec.	F1	Support
<i>Threshold = 0.50</i>				
No Pneumonia	0.00	0.00	0.00	1,419
Pneumonia	0.39	1.00	0.56	902
Accuracy			0.39	
<i>Threshold = 0.6460 (optimal F1)</i>				
No Pneumonia	1.00	0.00	0.00	1,419
Pneumonia	0.39	1.00	0.56	902
Accuracy			0.39	
ROC-AUC				0.5198

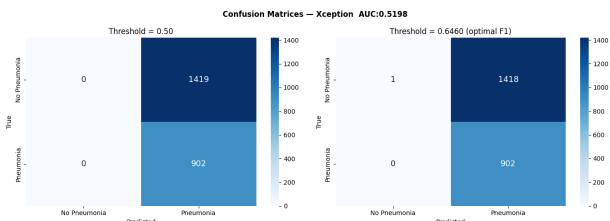


Figure 11: Xception confusion matrices. The model predicts all samples as Pneumonia at both thresholds. AUC = 0.5198.

Grad-CAM on Non-Converged Model. Despite training failure, Grad-CAM still produces spatially structured

heatmaps (Fig. 12), confirming that the XAI pipeline itself is technically functional. A false positive (Fig. 13) shows diffuse bilateral activation spanning the entire lung field—the model broadly activates on any radiograph, consistent with its degenerate all-positive prediction strategy.

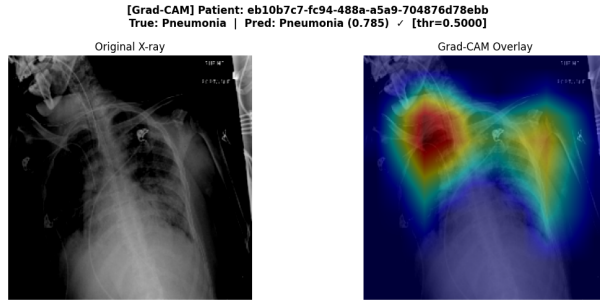


Figure 12: Xception Grad-CAM: true positive (prob=0.785). Despite non-convergence, the heatmap highlights bilateral peri-hilar regions.

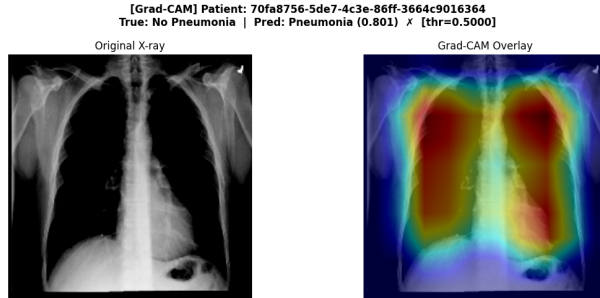


Figure 13: Xception Grad-CAM: false positive (prob=0.801). Diffuse bilateral activation reflects the model’s degenerate all-positive prediction strategy.

LIME. Figs. 14 and 15 show LIME explanations for a pneumonia-positive and a pneumonia-negative patient, respectively. For the positive case, LIME identifies left-sided opacity and mediastinal regions as contributing features. For the negative case correctly classified (prob=0.157), LIME highlights the cardiac silhouette and lower-lobe areas as positive features—despite a low overall probability—demonstrating that LIME provides locally faithful explanations even when the model’s global discrimination is poor.

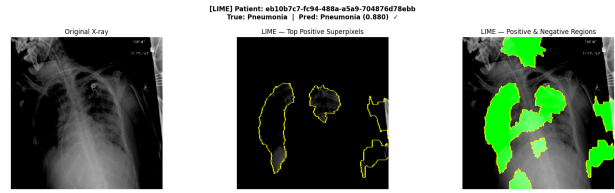


Figure 14: Xception LIME: true positive (prob=0.880). Positive superpixels align with left-sided opacity and hilar region.

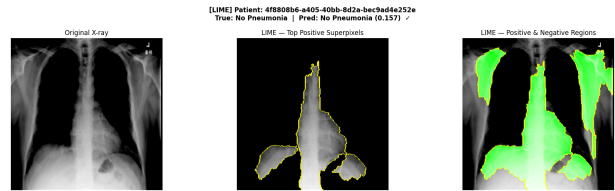


Figure 15: Xception LIME: true negative (prob=0.157). LIME identifies lung parenchyma and cardiac silhouette as positive features suppressed by the correct low-probability prediction.

4.4 Cross-Model Comparison

Table 7 provides a unified comparison across all three models and XAI metrics.

Table 7: Cross-Model Performance & XAI Summary

Metric	CheXNet	EffNet-B4	Xception
ROC-AUC	0.8687	0.8948	0.5198
Avg Precision	0.6699	0.8281	N/A
F1 (opt. thr.)	0.64	0.77	0.56
Opt. Threshold	0.5381	0.4243	0.6460
Epochs Ran	5	15	6
Pointing Game	70%	0%	N/A
Mean CAM Ratio	2.24	0.49	N/A
LIME Alignment	Good	Good	Partial
SHAP Magnitude	Med.	Low	Low
IG Faithfulness	High	High	Low

5 Discussion

5.1 Classification vs. Spatial Explainability

The most important finding of this study is the *dissociation between classification performance and spatial XAI faithfulness*. EfficientNet-B4 achieves the highest ROC-AUC (0.8948) yet produces Grad-CAM explanations that are completely spatially misaligned (0% Pointing Game, CAM ratio=0.49). This demonstrates that conventional

accuracy metrics are insufficient to characterise an AI system’s suitability for clinical deployment, and that XAI evaluation is a necessary—not optional—component of model validation.

CheXNet, despite lower classification performance ($AUC = 0.8687$), provides the most clinically interpretable Grad-CAM explanations (70% Pointing Game, CAM ratio = 2.24). The DenseNet-121 backbone’s dense connectivity facilitates the propagation of spatial information through the network, yielding feature maps whose gradients more faithfully reflect lesion location. This is consistent with the original CheXNet paper [1], which noted that CAM localisation approximates radiologist annotations.

5.2 EfficientNet-B4 Grad-CAM Failure

The degenerate peak activation at (364, 0) for all EfficientNet-B4 patients is a systematic artefact, not a random error. Compound scaling reduces the spatial feature map from 380×380 to 12×12 , and the aggressive depthwise convolution stacking in MBConv blocks appears to concentrate gradient flow toward the feature map boundary, where zero-padding artefacts accumulate. This failure mode is architecture-specific and would not be detectable by AUC alone.

Critically, LIME explanations for the same EfficientNet-B4 model (Fig. 8) are clinically meaningful, correctly identifying opacity regions. This demonstrates that model-agnostic perturbation-based methods are more robust to architecture-specific gradient pathologies than gradient-based methods like Grad-CAM. For architectures with aggressive spatial downsampling, LIME or IG should be preferred over Grad-CAM.

5.3 Xception Training Failure

Xception’s failure to converge ($AUC = 0.52$) is attributable to multiple compounding factors. First, the dataset used for Xception (full 26,684 patients, positive rate 22.5%) differs from those used for CheXNet and EfficientNet-B4 (balanced subsets), making direct comparison invalid. Second, AMP was disabled for Xception (due to numerical instability in depthwise separable convolutions), slowing training. Third, the epoch limit of 6 (vs. 15 for EfficientNet-B4) likely prevented sufficient fine-tuning after backbone unfreezing. The wildly oscillating validation loss ($5.77 \rightarrow 6.19 \rightarrow 4.65 \rightarrow 1.61 \rightarrow 2.93 \rightarrow 5.59$) suggests the learning rate of $5e-5$ was too high after backbone unfreezing for this architecture.

5.4 SHAP Limitations

SHAP attributions for EfficientNet-B4 show low magnitude (± 0.04), reflecting the minimal background set

(2–4 images) used for the GradientExplainer. A more robust evaluation would use 50–100 background images and 100+ gradient samples. Despite low magnitude, SHAP attributions are directionally consistent with the model’s prediction, confirming that the theoretical grounding of Shapley values provides meaningful—if low-resolution—explanations even under computational constraints.

5.5 Clinical Implications

From a deployment perspective, CheXNet offers the best combination of classification accuracy ($AUC = 0.87$) and XAI faithfulness (Pointing Game = 70%, CAM ratio = 2.24) for a clinical triage application. LIME should be used as a complementary or primary explanation for architectures with aggressive spatial downsampling (EfficientNet, MobileNet families).

Threshold calibration is clinically critical: the default 0.50 threshold produced suboptimal recall (73%) for CheXNet, while the F1-optimal threshold of 0.5381 improved recall to 70% with higher precision. In emergency radiology triage, a lower threshold (higher recall) is preferable to minimise missed pneumonia, and the optimal threshold should be calibrated on the target clinical population.

5.6 Comparison with Prior Work

Our CheXNet Pointing Game results (70%) are consistent with the qualitative localisation alignment reported by Rajpurkar et al. [1] and quantitative evaluations in the broader XAI-in-radiology literature. Our finding that LIME is more robust than Grad-CAM for architectures with high spatial compression corroborates Holzinger et al.’s [12] recommendation for multi-method XAI evaluation in clinical AI systems.

6 Conclusion

This paper presented a comprehensive comparative study of three CNN architectures and four XAI techniques for binary pneumonia detection from chest radiographs using the RSNA Pneumonia Detection Challenge dataset.

The key findings are:

1. **EfficientNet-B4 achieves the highest classification performance** ($AUC = 0.8948$, $AP = 0.8281$), but its Grad-CAM explanations are clinically meaningless due to architecture-specific spatial degeneration (0% Pointing Game).
2. **CheXNet provides the best XAI-classification balance** ($AUC = 0.8687$, Pointing Game = 70%, CAM ra-

tio = 2.24), making it the most suitable architecture for clinically deployed explainable pneumonia detection.

3. **Xception failed to converge** (AUC = 0.52) under the experimental protocol applied, establishing a performance lower bound and illustrating the importance of dataset consistency and training stability across architectures.
4. **LIME is more robust than Grad-CAM** for architectures with aggressive spatial downsampling. SHAP and IG provide complementary pixel-level perspectives with distinct theoretical guarantees.
5. **High classification accuracy does not guarantee spatial XAI faithfulness**—a finding with direct implications for regulatory evaluation of AI-based medical imaging devices.

Future Work: (1) Re-training Xception with a more aggressive warm-up schedule and consistent dataset protocol; (2) exploring Grad-CAM++ and Score-CAM as alternatives for compound-scaled architectures; (3) clinical user studies evaluating whether XAI overlays improve radiologist confidence; (4) deploying the best-performing model (EfficientNet-B4 with LIME) in a prospective clinical trial under FDA SaMD guidance; (5) Vision Transformer architectures and attention map visualisations as a complementary XAI paradigm.

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